

Connectedness and Compactness

General Topology

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Definition

A topological space X is said to be **disconnected** if it can be expressed as the union of two disjoint, nonempty, open subsets. Any such subsets are said to **disconnect** X . If X is not disconnected, it is said to be **connected**.

Theorem 4.1

A topological space X is connected if and only if the only subsets of X that are both open and closed in X are $\emptyset$ and X itself.

Theorem 4.2

Suppose X is a nonempty connected space. Then every continuous map from X to a discrete space is constant.

Theorem 4.3 (Main Theorem on Connectedness)

Let X, Y be topological spaces and let $f : X \rightarrow Y$ be a continuous map. If X is connected, then $f(X)$ is connected.

Theorem 4.4 (Topological Invariance of Connectedness)

Every space homeomorphic to a connected set is connected.

Theorem 4.5 (Properties of Connected Spaces)

- i) Suppose X is any space and U, V are disjoint open subsets of X . If A is a connected subset of X contained in $U \cup V$, then either $A \subseteq U$ or $A \subseteq V$.
- ii) If X is a space that contains a dense connected subset, then X is connected.
- iii) Suppose X is any space and $A \subseteq X$ is connected. Then $\overline{A}$ is connected, as is any subset B such that $A \subseteq B \subseteq \overline{A}$.
- iv) Let X be a space, and let $\{B_\alpha\}_{\alpha \in I}$ be a collection of connected subspaces of X with a point in common. Then $\bigcup_{\alpha \in I} B_\alpha$ is connected.
- v) Every product of finitely many connected spaces is connected.
- vi) Every quotient space of a connected space is connected.

Theorem 4.6

A nonempty subset of $\mathbb{R}$ is connected if and only if it is a singleton or an interval.

Theorem 4.7 (Generalised Bolzano's Intermediate Value Theorem)

Suppose X is a connected topological space, and $f : X \rightarrow \mathbb{R}$ is continuous. If $p, q \in X$, then f attains every value between $f(p)$ and $f(q)$.

Definition

Let X be a topological space and $p, q \in X$. A **path in X from p to q** is a continuous map $f : I \rightarrow X$ such that $f(0) = p$ and $f(1) = q$, where $I := [0, 1]$ is the unit interval. We say that X is **path-connected** if for every $p, q \in X$, there is a path in X from p to q .

Lemma

A topological space X is path-connected if and only if for every $p, q \in X$ there is a continuous map $f : [a, b] \rightarrow X$ such that $f(a) = p$ and $f(b) = q$.

Theorem 4.8 (Properties of Path-Connected Spaces)

- i) *Every continuous image of a path-connected space is path-connected.*
- ii) *Let X be a space, and let $\{B_\alpha\}_{\alpha \in A}$ be a collection of path-connected subspaces of X with a point in common. Then $\bigcup_{\alpha \in A} B_\alpha$ is path-connected.*
- iii) *Every product of finitely many path-connected spaces is path-connected.*
- iv) *Every quotient space of a path-connected space is path-connected.*

Theorem 4.9

Path connectedness implies connectedness.

Example 4.10

The following spaces are path-connected, and therefore connected.

- a) $\mathbb{R}^n$.
- b) Any subset $B \subseteq \mathbb{R}^n$ that is **convex**, which means that for any $x, x' \in B$, the line segment from x to x' lies entirely in B .
- c) $\mathbb{R}^n \setminus \{0\}$ for $n \geq 2$.
- d) $\mathbb{S}^n$ for $n \geq 1$.
- e) The n -torus $\mathbb{T}^n$.

Example 4.11

Define the subsets of the plane by

$$T_0 = \{(x, y) : x = 0 \text{ and } y \in [-1, 1]\};$$

$$T_+ = \{(x, y) : x \in (0, 2/\pi) \text{ and } y = \sin(1/x)\}.$$

Let $T = T_0 \cup T_+$. The space is called the **topologist's sine curve**. This is an example of a connected space that is not path connected.

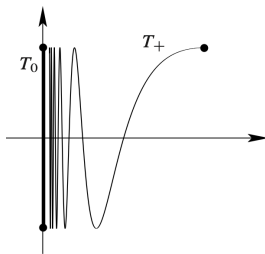


Figure 1: The topologist's sine curve

Definition

Let X be a topological space. A **component of X** is a maximal nonempty connected subset of X , that is, a nonempty connected subset that is not properly contained in any other connected subset.

Theorem 4.12

If X is any topological space, its components form a partition of X .

Theorem 4.13

A nonempty topological space is connected if and only if it has exactly one component.

Example 4.14

- *The components of $\mathbb{R} \setminus \{0\}$ are the positive half and the negative half-line.*
- *The components of the union of two disjoint closed discs are the two discs themselves.*

Theorem 4.15 (Properties of Components)

Let X be a nonempty topological space.

- i) Each component of X is closed in X .*
- ii) Any nonempty connected subset of X is contained in a single component.*

Definition

If X is any space, define a **path component of X** to be a maximal nonempty path-connected subset.

Theorem 4.16

A space is path-connected if and only if it has exactly one path component.

Theorem 4.17 (Properties of Path Components)

Let X be any space.

- i) The path components of X form a partition of X .*
- ii) Each path component is contained in a single component, and each component is a disjoint union of path components.*
- iii) Any nonempty path-connected subset of X is contained in a single path component.*

Definition

We say that a space is **locally connected** if it admits a basis of connected open subsets, and **locally path-connected** if it admits a basis of path-connected open subsets.

Theorem 4.18

Every manifold (with or without boundary) is locally connected and locally path-connected.

Theorem 4.19 (Properties of Locally Connected Spaces)

Suppose X is a locally connected space.

- i) Every open subset of X is locally connected.*
- ii) Every component of X is open.*

Theorem 4.20 (Properties of Locally Path-Connected Spaces)

Suppose X is a locally path-connected space.

- i) X is locally connected.*
- ii) Every open subset of X is locally path-connected.*
- iii) Every path component of X is open.*
- iv) The path components of X are equal to its components.*
- v) X is connected if and only if it is path-connected.*

Definition

An **open cover** of a space X is a collection $\mathcal{U}$ of open subsets of X whose union is X , and a **subcover** of $\mathcal{U}$ is a subcollection of elements of $\mathcal{U}$ that still covers X . A topological space X is said to be **compact** if every open cover of X has a finite subcover; or in other words, if given any open cover $\mathcal{U}$ of X , there are finitely many sets $U_1, \dots, U_k \in \mathcal{U}$ such that

$$X = \bigcup_{\alpha=1}^k U_{\alpha}.$$

A subset $S \subseteq X$ is said to be **compact** if it is compact when endowed with the subspace topology.

Definition

*If X is a topological space and $A \subseteq X$, a collection of subsets of X whose union contains A is also called a **cover** of A ; if the subsets are open in X we sometimes call it an **open cover** of A . A collection of subsets of A whose union is A is called a **relative cover** of A ; if the subsets are open in A , we call it a **relatively open cover** of A .*

Theorem 4.21 (Compactness Criterion for Subspaces)

If X is any topological space, a subset $A \subseteq X$ is compact (in the subspace topology) if and only if every cover of A by open subsets of X has a finite subcover.

Theorem 4.22

Suppose X is a topological space, and (x_n) a sequence of points in X converging to $x \in X$. Then the set $A = \{x_n : n \in \mathbb{N}\} \cup \{x\}$ is compact.

Theorem 4.23 (Main Theorem on Compactness)

Let X and Y be topological spaces, and let $f : X \rightarrow Y$ be a continuous map. If X is compact, then $f(X)$ is compact.

Theorem 4.24 (Topological Invariance of Compactness)

Every space homeomorphic to a compact space is compact.

Theorem 4.25

If X is a Hausdorff space and $A, B \subseteq X$ are disjoint compact subsets, there exist disjoint open subsets $U, V \subseteq X$ such that $A \subseteq U$ and $B \subseteq V$.

Lemma 4.26 (Tube Lemma)

Let X be any space and let Y be a compact space. If $x \in X$ and $U \subseteq X \times Y$ is an open subset containing $\{x\} \times Y$, then there is a neighbourhood V of x in X such that $V \times Y \subseteq U$.

Theorem 4.27 (Properties of Compact Spaces)

- i) *Every closed subset of a compact space is compact.*
- ii) *Every compact subset of a Hausdorff space is closed.*
- iii) *Every compact subset of a metric space is bounded.*
- iv) *Every finite product of compact spaces is compact.*
- v) *Every quotient of a compact space is compact.*

Theorem 4.28 (Heine-Borel)

The compact subsets of $\mathbb{R}^n$ are exactly the closed and bounded ones.

Theorem 4.29 (Generalised Weierstrass Extreme Value Theorem)

If X is a compact space and $f : X \rightarrow \mathbb{R}$ is continuous, then f is bounded and attains its maximum and minimum values on X .

Definition

A space X is said to be **limit point compact** if every infinite subset of X has a limit point in X , and **sequentially compact** if every sequence of points in X has a subsequence that converges to a point in X .

Theorem 4.30

Compactness implies limit point compactness.

Theorem 4.31

For first countable Hausdorff spaces, limit point compactness implies sequential compactness.

Theorem 4.32

For metric spaces and second countable topological spaces, sequential compactness implies compactness.

Corollary

For metric spaces and second countable Hausdorff spaces, limit point compactness, sequential compactness, and compactness are all equivalent properties.

Theorem 4.33 (Bolzano-Weierstrass)

Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Theorem 4.34

Endowed with the Euclidean metric, a subset of $\mathbb{R}^n$ is a complete metric space if and only if it is closed in $\mathbb{R}^n$. In particular, $\mathbb{R}^n$ is complete.

Theorem 4.35

Every compact metric space is complete.

Definition

A topological space X is said to be **locally compact** if for every $p \in X$ there is a compact subset of X containing a neighbourhood of p . A subset A of X is said to be **precompact** (or sometimes **relatively compact**) in X if $\overline{A}$ is compact.

Theorem 4.36

Let X be a Hausdorff space. Then the following are equivalent:

- i) X is locally compact.*
- ii) Each point of X has a precompact neighbourhood.*
- iii) X has a basis of precompact open subsets.*

Theorem 4.37

Every manifold with or without boundary is locally compact.

Theorem 4.38

Let X be a locally compact Hausdorff space. If $x \in X$ and U is any neighbourhood of x , there exists a precompact neighbourhood V of x such that $\overline{V} \subseteq U$.

Theorem 4.39

Any open or closed subset of a locally compact Hausdorff space is a locally compact Hausdorff space.